Two matchings with equal characteristic value admit representatives meeting exactly in the certificates

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

arxiv:2210.00245v2; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's labelled environment at source0.tex lines 475-477 and its proof environment opens at line 478; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. Preamble: the author's own package lines and theorem-environment family (source lines 5-24); the layout packages of the source are not retained because the retained material does not use them.

Editorial note. Retained-licence form: the licence evidence of this source carries no acceptance record, and none may be written on anyone's behalf, so the card points at a task-directory evidence copy in the team-wide four-key reuse form (name, version_url, prior_license_path, basis) whose basis states that the source's staged record is reused as a material basis only. No status key and no acceptance record are written.

Editorial note. Metadata provenance: source_version is the value recorded in the source's licence evidence (2210.00245v2); the author list is transcribed from the authors_index key of the source record of the pinned copy, and the title from its title key.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; the identity of the counted item is its author label.

Editorial note. r002 note: r001 failed with an undefined control sequence because the counted statement uses the author's macro for the family of perfect matchings, and the author's macro block (source lines 26-29) had not been retained; r002 adds exactly that block as a further preamble span and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: representatives meeting exactly in the certificates for the perfect-matching case (author label lem:2-int-coset-pms).

Allowed standard prerequisites (editorial).

- 2-intersecting family of perfect matchings. A family of perfect matchings such that any two of its members share at least two pairs.. Its characteristic function is the object of the statement.

- Certificate of a matching. The set of pairs common to the members of the family, used to name them.. They are the sets whose intersection is prescribed.
- Size bound. The common certificate size is at most n minus two, so at least four indices remain free.. It is what makes the explicit pairing possible.

Lemma 0.1. *Let $f\colon \mathcal{M}_{2n} \to \{0,1\}$ be the characteristic function of a 2-intersecting family, and suppose that $C(f) \leq n-2$. If $f(m_1) = f(m_2) = 1$ and C_{m_1}, C_{m_2} are minimum certificates for m_1, m_2 (respectively), then $|C_{m_1} \cap C_{m_2}| \geq 2$.*

Proof. We will show how to construct a perfect matching m_1' satisfying C_{m_1} such that $m_1' \cap C_{m_2} = C_{m_1} \cap C_{m_2}$. The same argument can be used to construct a perfect matching m_2' satisfying C_{m_2} such that $m_1' \cap m_2' = m_1' \cap C_{m_2}$. It follows that $|C_{m_1} \cap C_{m_2}| = |m_1' \cap m_2'| \geq 2$, since f is the characteristic function of a 2-intersecting family.

Let $C_{m_1} = \big\{\{i_1, j_1\}, \ldots, \{i_r, j_r\}\big\}$, where $r \leq n-2$, and let $k_1, \ldots, k_{2(n-r)}$ be the indices not mentioned in C_{m_1}. Rearrange the indices so that if $\{k_s, k_t\} \in C_{m_2}$ then $\{s,t\} = \{2\ell - 1, 2\ell\}$ for some $\ell \in [n-r]$. The perfect matching m_1' has the pair representation

$$\{i_1, j_1\}, \ldots, \{i_r, j_r\}, \{k_1, k_{n-r+1}\}, \{k_2, k_{n-r+2}\}, \ldots, \{k_{n-r}, k_{2(n-r)}\}.$$

Since $n - r \geq 2$, the new pairs do not intersect C_{m_2}, completing the proof. □